

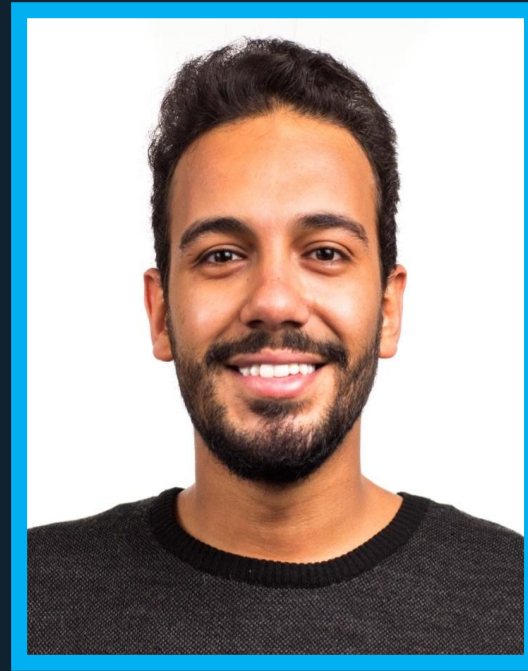
ITI AI-Pro

Natural Language Processing



RULES

Acknowledgment



CONTENTS

- **Intro to NLP**
- **Text Preprocessing**
- **Traditional Text Representation**
- **Traditional NLP**
- **Advanced Text Representation**
 - Word Embeddings
 - word2vec
 - GloVe

Part I

Intro to NLP



DEFINITION

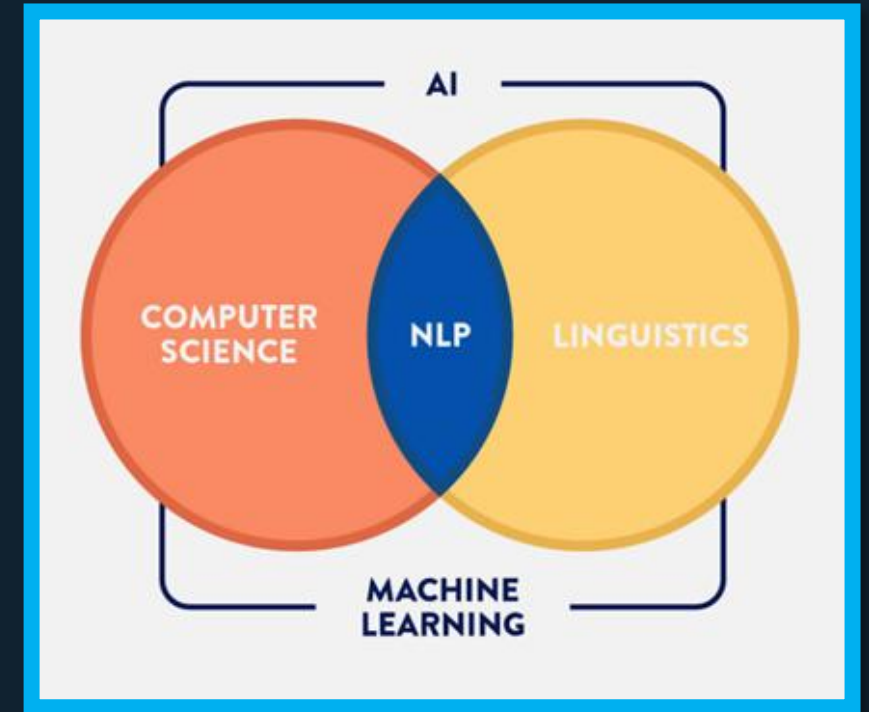


**What is
NLP?**

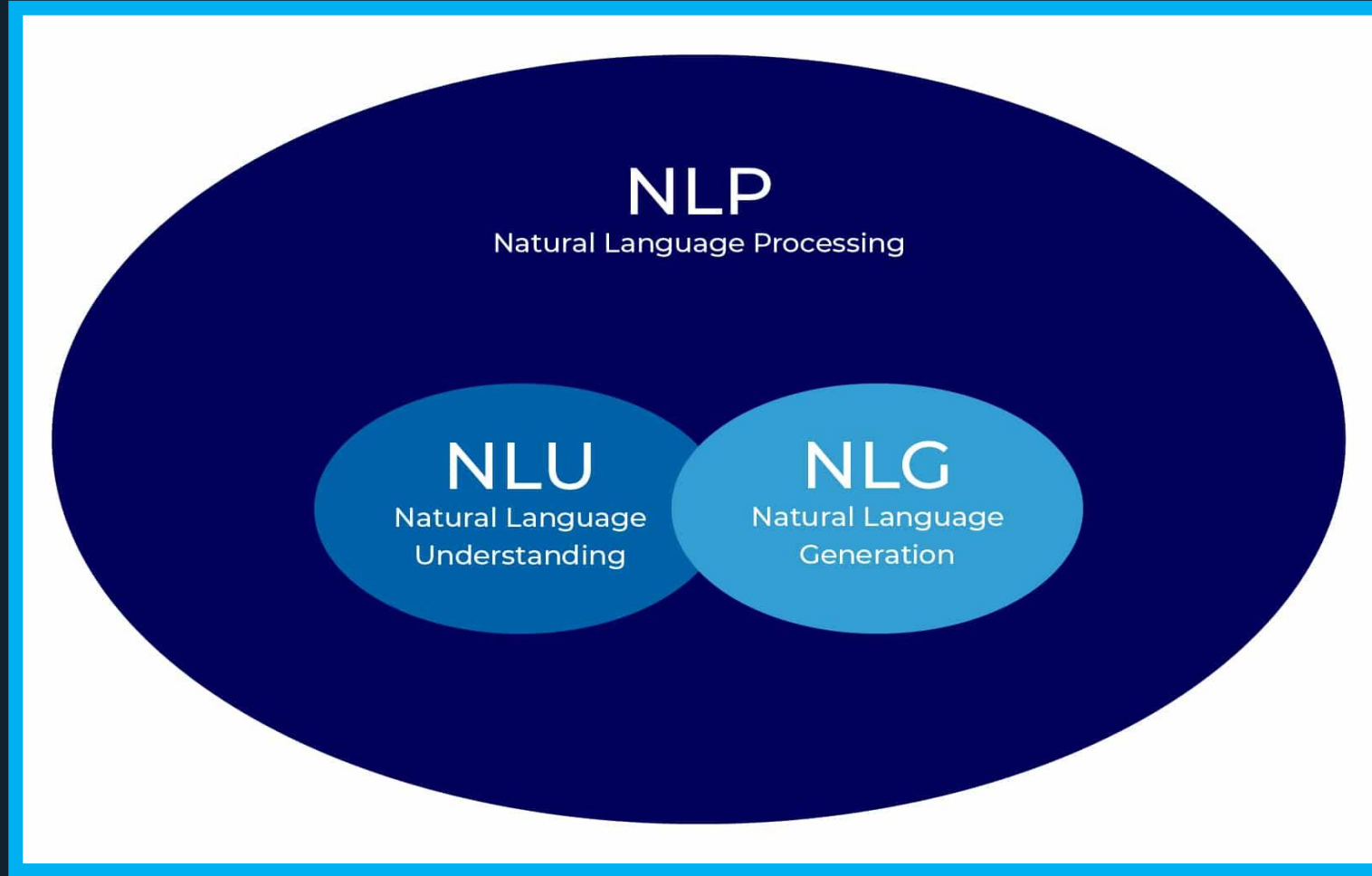


DEFINITION

“
Branch of Science merges
CS, AI and Linguistics
so that machines can
understand and interact
with Human Languages”



DEFINITION



DEFINITION

NLG

- Discourse Generation
- Lexical Choice
- Sentence Planning/ Generation
- Realization
- Document Structuring

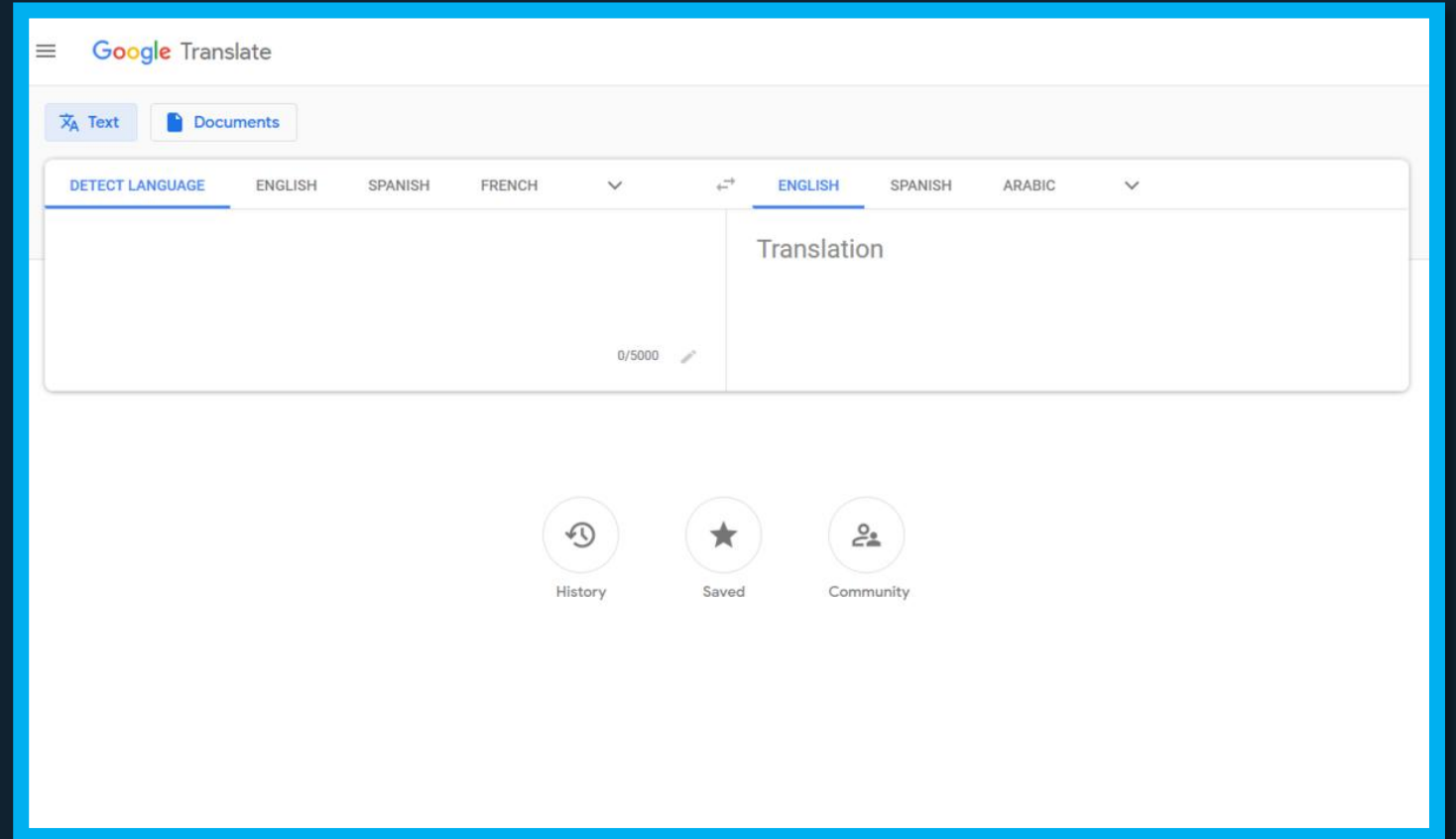
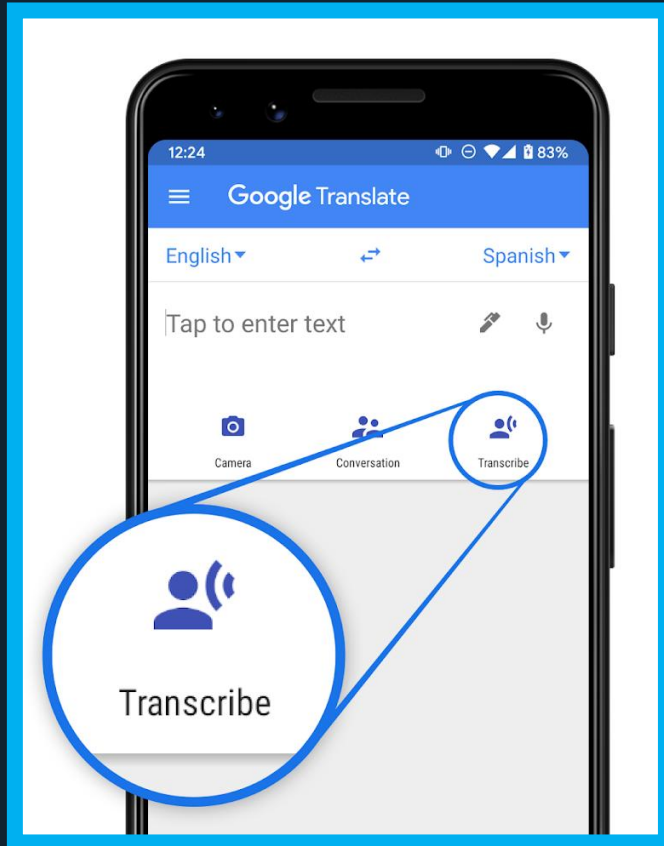
NLP

- Named entity Recognition (NER)
- Part-of-Speech Tagging (POS)
- Syntactic Parsing
- Coreference Resolution
- Machine Translation

NLU

- Lexical Ambiguity/ Analysis
- Sentiment Analysis
- Topic Classification
- Information Extraction
- Entity Detection
- Summarization
- Semantic Parsing
- Syntactical Analysis

APPLICATIONS



APPLICATIONS



APPLICATIONS



APPLICATIONS



LLaMA
 Meta

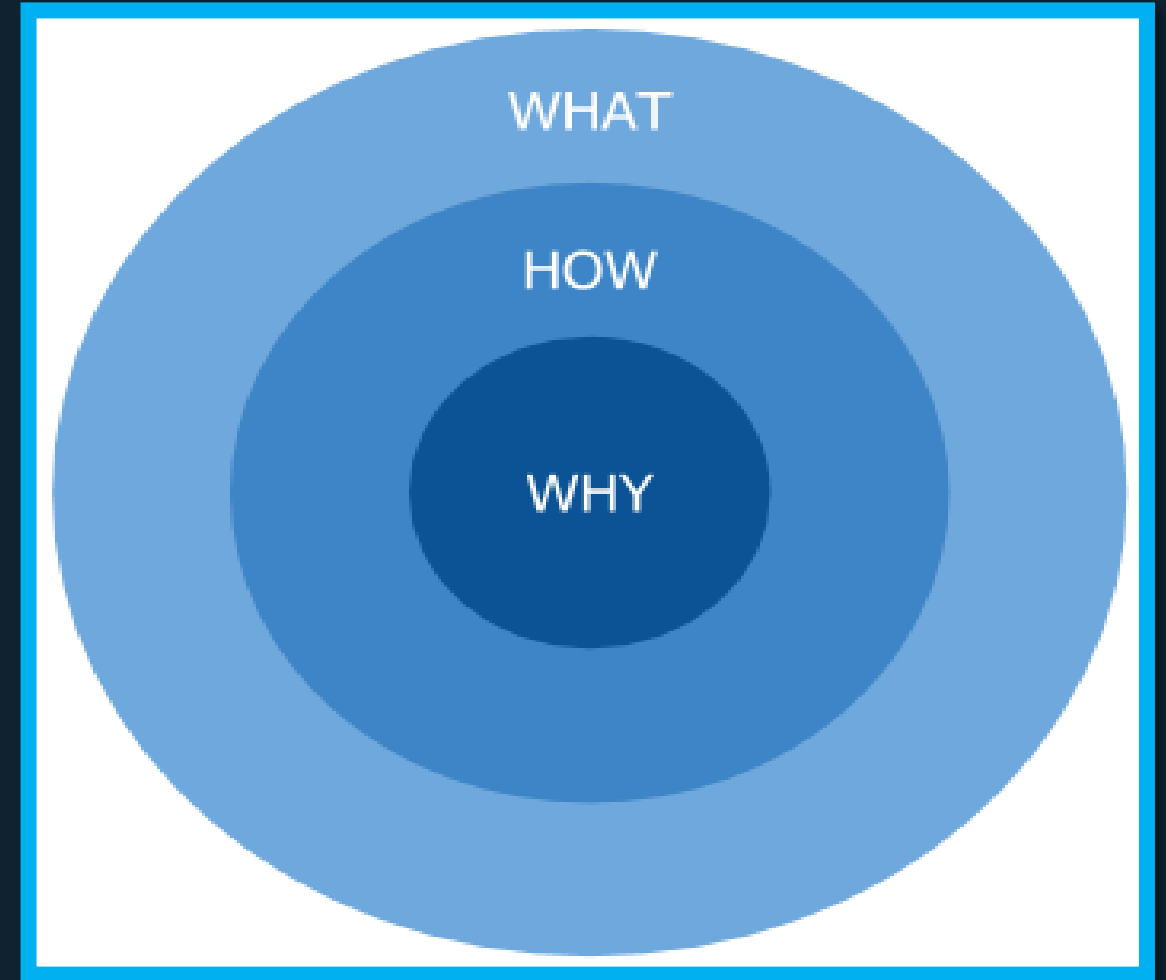
Part II

Text Preprocessing



TEXT PREPROCESSING

- Lowercasing
- Remove URLs
- Remove Punctuations
- Remove Stopwords
- Stemming
- Lemmatization
- ...
- ...



Valuable Source

NB I RegEx



NB II
Text
Normalization



NB III
Edit
Distance



NB IV
Text
Preprocessing



Part III

Text

Representation



Words

VS

Numbers

BOW

➤ Pros:

- Simple

➤ Cons:

- Independent Words
- No Order

	cat	dog	eats	food	red	the
the red dog	0	1	0	0	1	1
cat eats dog	1	1	1	0	0	0
dog eats food	0	1	1	1	0	0
red cat eats	1	0	1	0	1	0

BOW N-Grams

➤ Pros:

- Context

➤ Cons:

- High Dimensions

	cat	cat eats	dog	dog eats	eats	eats dog	eats food	food	red	red cat	red dog	the	the red
the red dog	0	0	1	0	0	0	0	0	1	0	1	1	1
cat eats dog	1	1	1	0	1	1	0	0	0	0	0	0	0
dog eats food	0	0	1	1	1	0	1	1	0	0	0	0	0
red cat eats	1	1	0	0	1	0	0	0	1	1	0	0	0

TF - IDF

➤ Pros:

- Unique Words

➤ Cons:

- Same as BOW

$$w_{i,j} = tf_{i,j} \times \log \left(\frac{N}{df_i} \right)$$

	cat	dog	eats	food	hot	red	the
the red dog	0.000000	0.442740	0.000000	0.000000	0.000000	0.634027	0.634027
cat eats dog	0.677803	0.473309	0.562638	0.000000	0.000000	0.000000	0.000000
dog eats food	0.000000	0.423954	0.503968	0.752515	0.000000	0.000000	0.000000
red cat eats	0.609818	0.000000	0.506204	0.000000	0.000000	0.609818	0.000000
the hot dog	0.000000	0.401565	0.000000	0.000000	0.712775	0.000000	0.575063

NB V

**Features
Extraction**



NB VI
Text
Classification



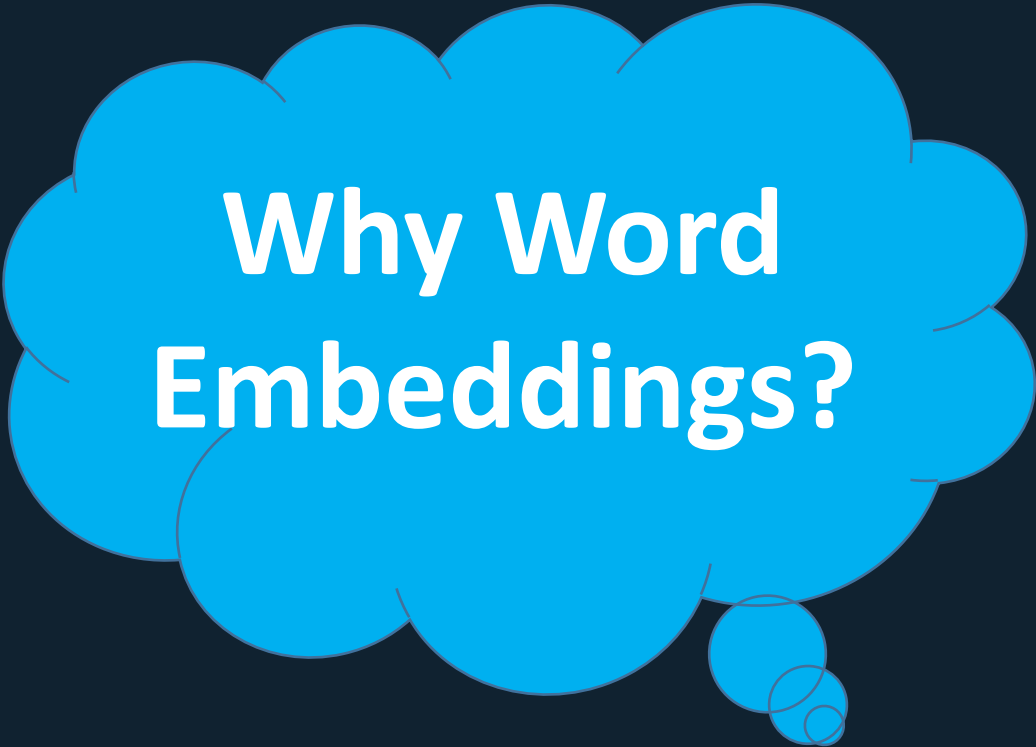
Part V

Advanced

Text Representation



WORD EMBEDDING



Why Word
Embeddings?

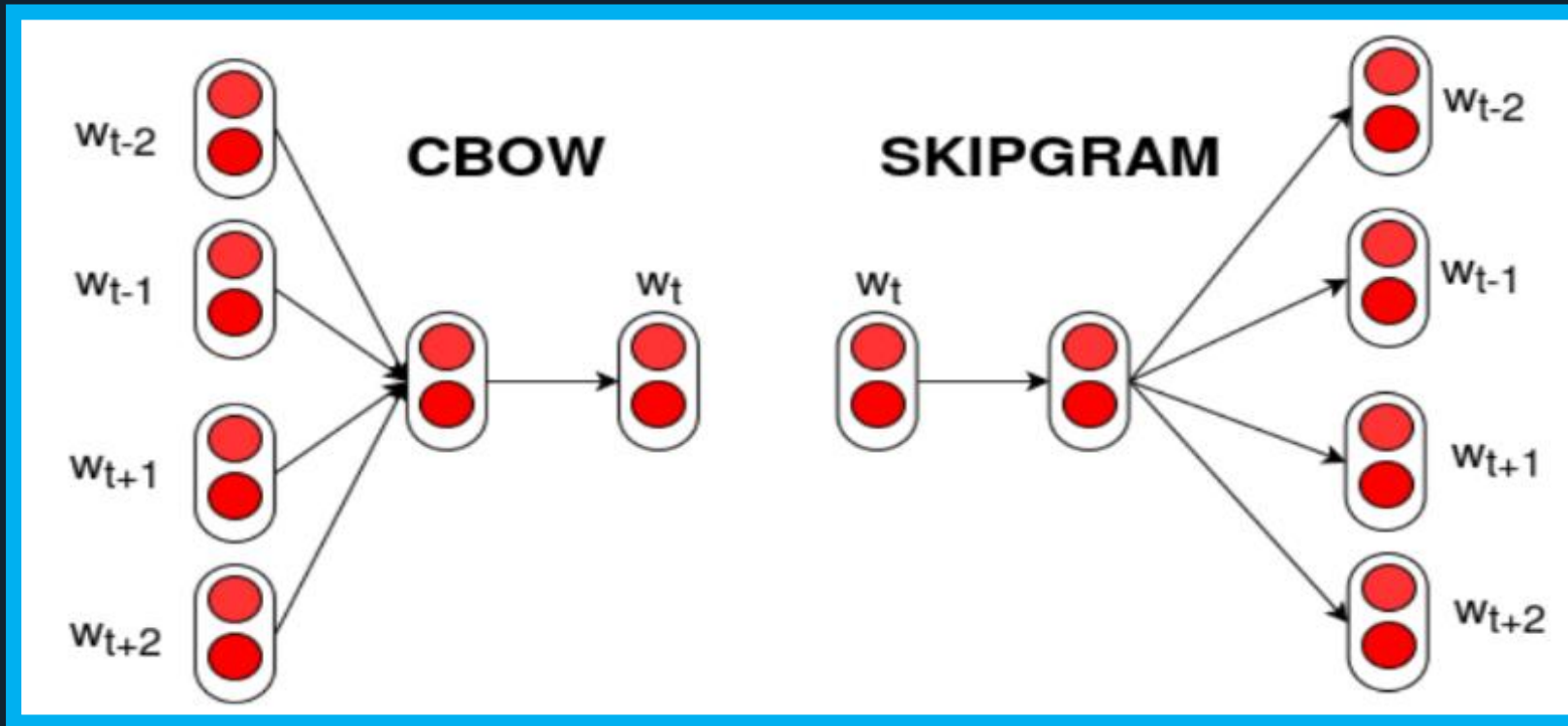


WORD EMBEDDING

	Bag-of-Words	Word Embeddings
Simplicity		
Contextual Info.		
Dimensions	No. of Sentences	Any number

WORD 2 VEC

- Train NN **given** words / context
- NN **predicts** the context / word



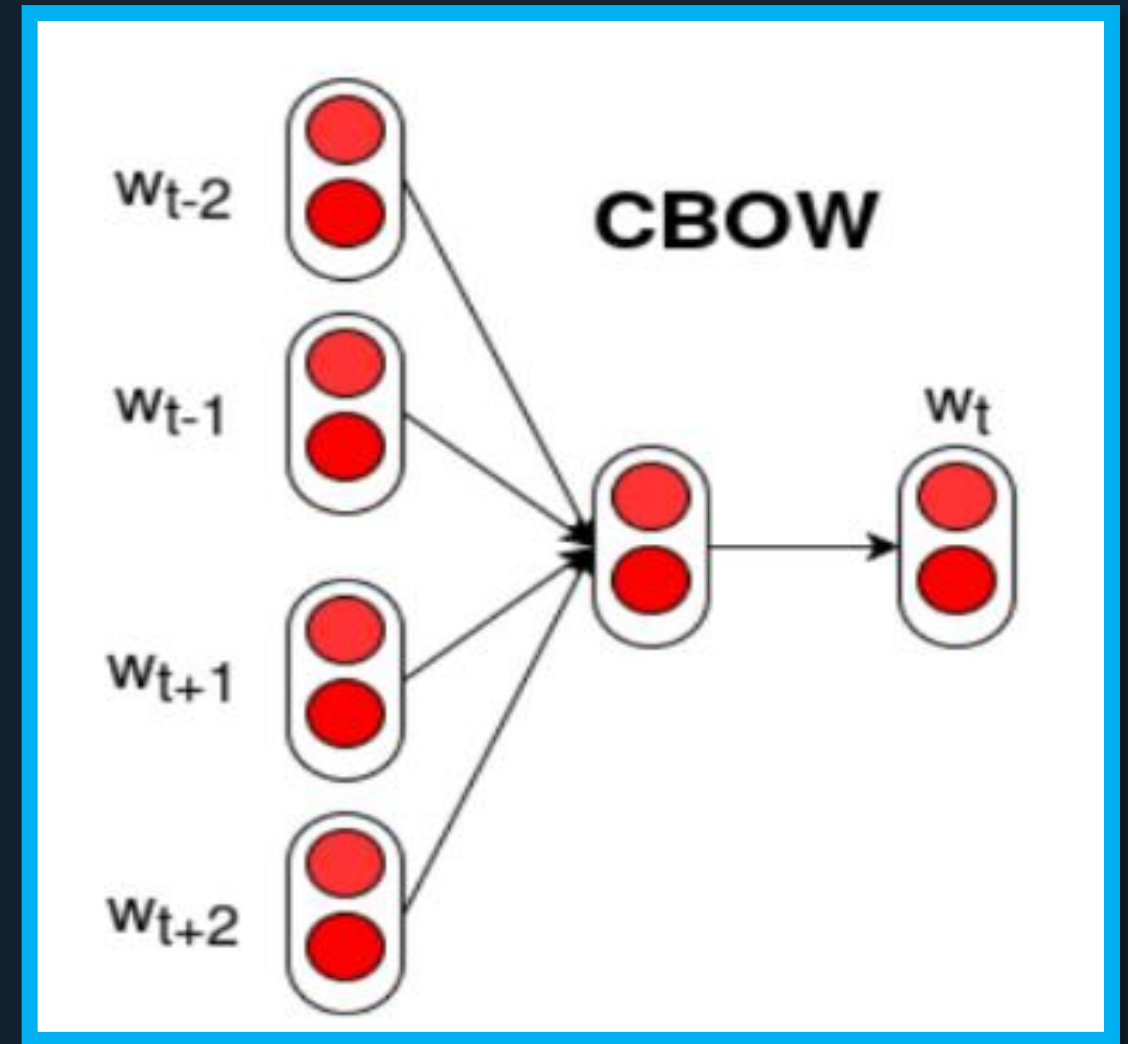
WORD 2 VEC

- **CBOW**

Continuous Bag of Words

- **Input** : Context Words
- **Output** : Center Word

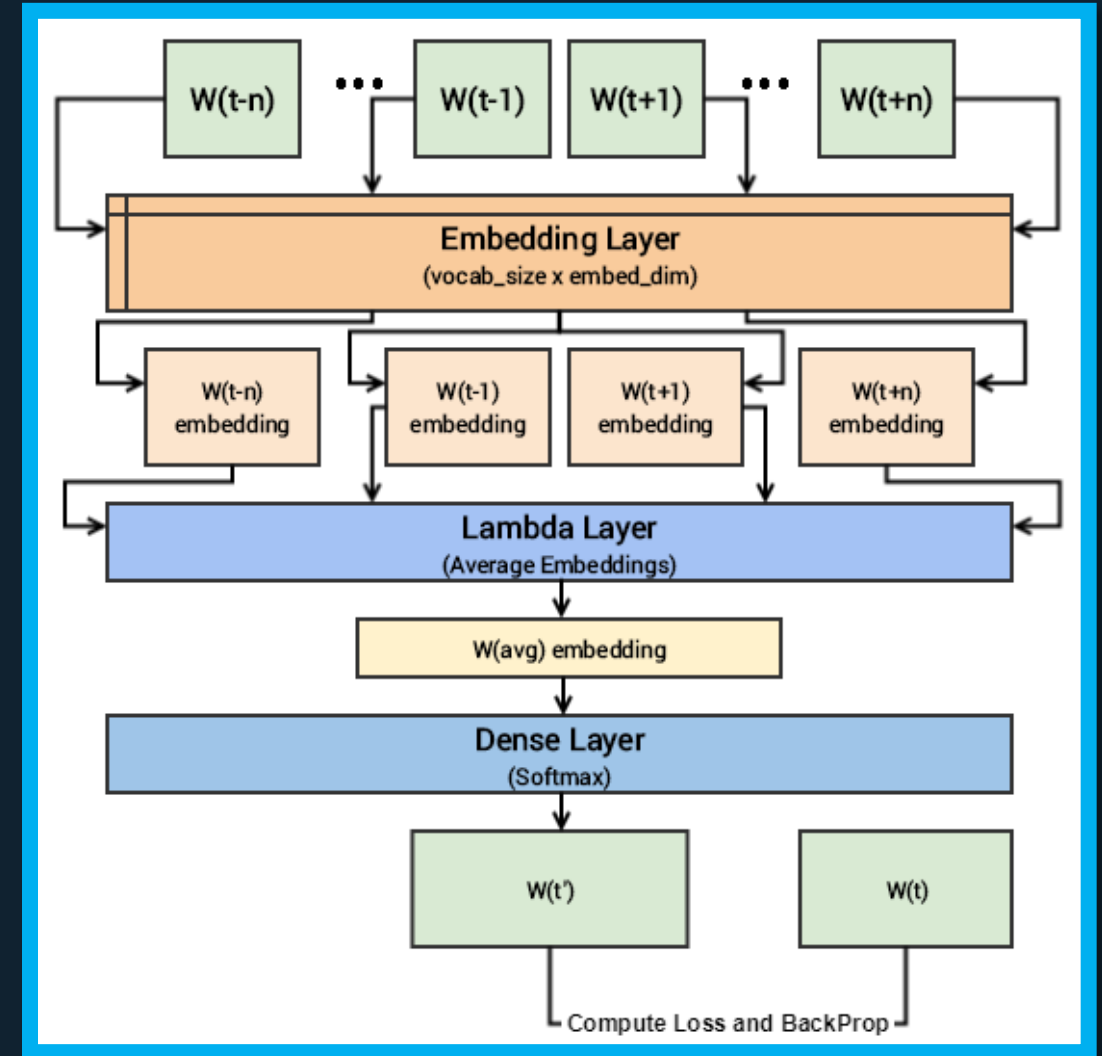
[More](#)



WORD 2 VEC

Steps:

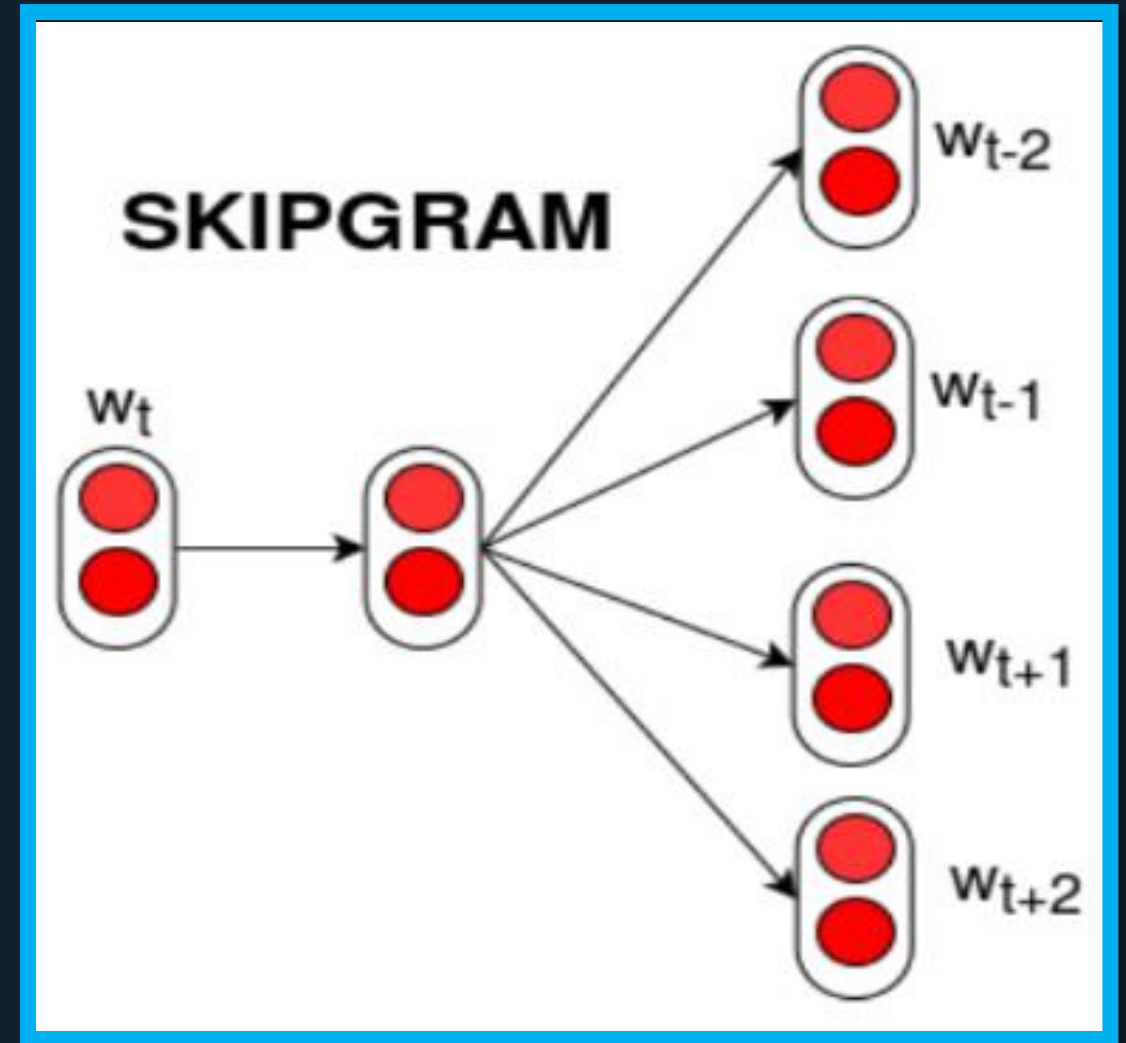
1. **Feed** context words to same embedding layer
2. **Average** embeddings
3. **Feed** the average to dense layer that tries to predict the word
4. **Compare** predicted and true words
5. **Back** propagate



WORD 2 VEC

- Skip - Gram

- Input : Word
- Output : Context

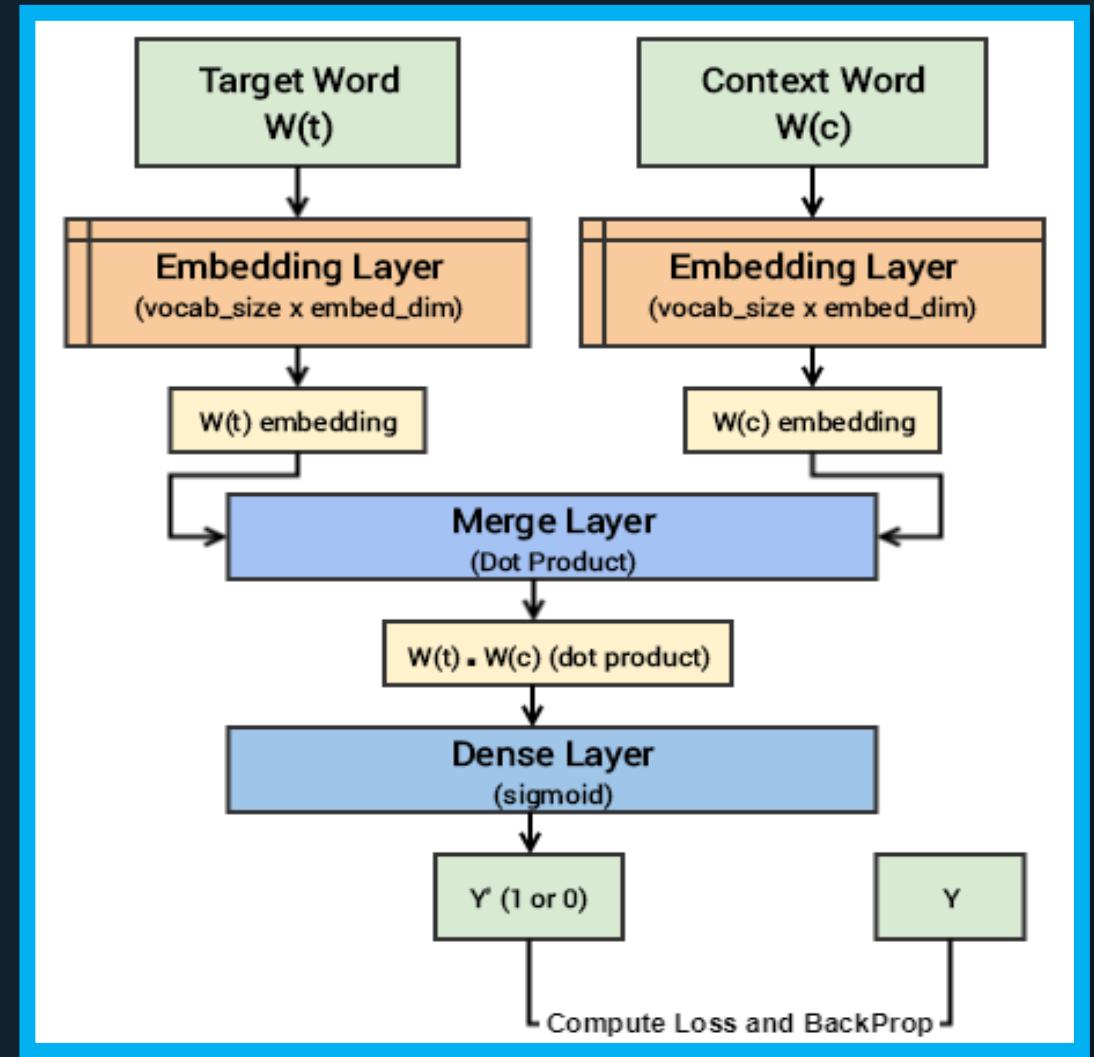


[More](#)

WORD 2 VEC

Steps:

1. **Given** word and context
2. **Generate** embeddings
3. **Merge** and **Feed** to dense layer
4. Sigmoid layer **predict** if similar or not
5. **Back** propagate



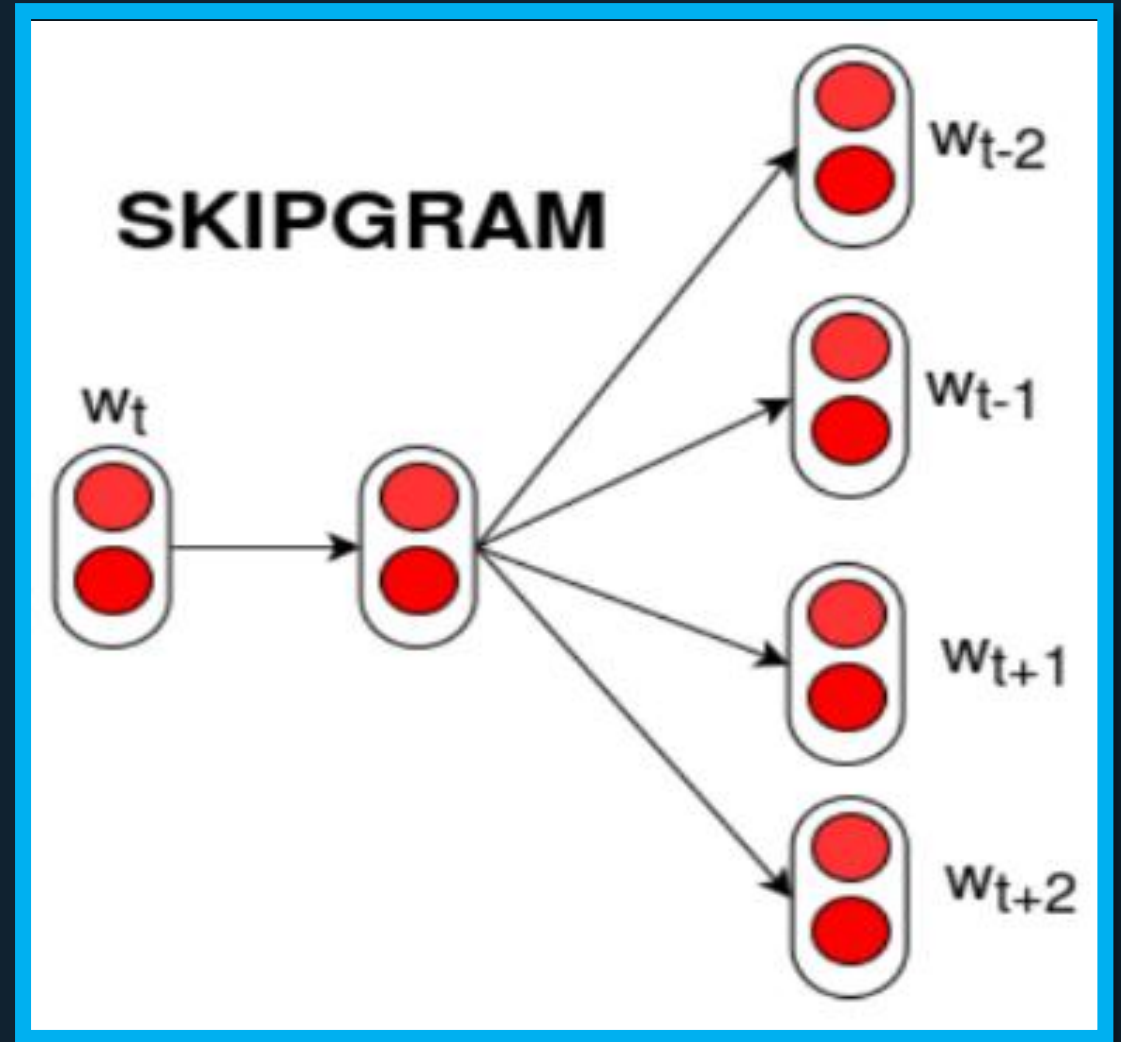
WORD 2 VEC

- Skip - Gram

Negative Sampling

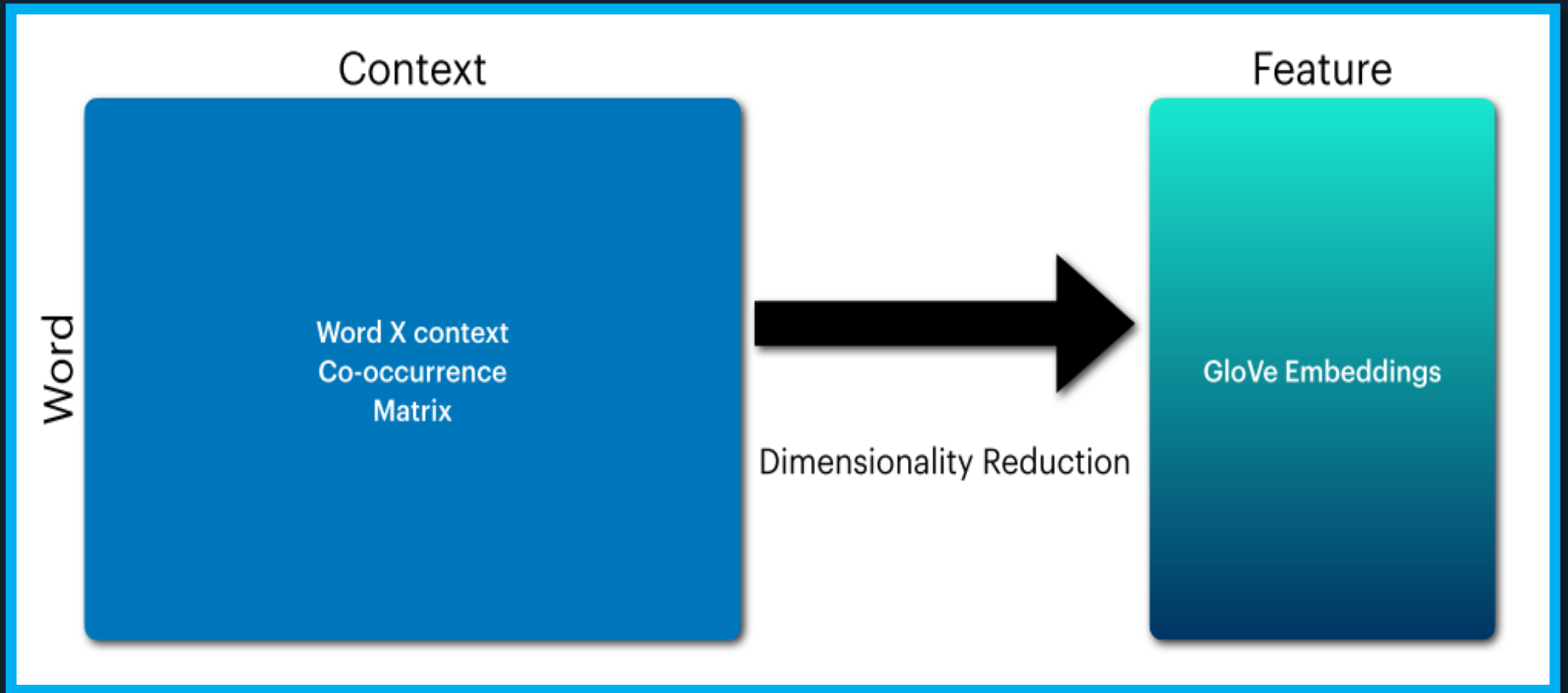
(word, context, 0)

(word, context, 1)





GLOVE



GLOVE VS WORD 2 VEC

	GloVe	Word 2 Vec
Contextual Info.		
Performance	Same	Same
Criteria	Count-based Dimensionality Reduction	NN

[More](#)



THANKS
A LOT